Business Intelligence Internship

Data Simulation & Analytics Report

Car Insurance Portfolio | Policy Year 2024 | Claims 2025–2026

Prepared by: Soumodwip Mondal

1,000,000	49,297	INR 23.9 Cr	INR 49.45 Cr
Total Policies	Total Claims Filed	Total Premium	Total Claims Cost

1. Executive Summary

This report presents the findings of a car insurance portfolio simulation for the year 2024, covering 1,000,000 policies sold across the calendar year. The analysis spans policy data creation, claims simulation for 2025 and the first two months of 2026, and a comprehensive set of business analytics including premium collection, claim cost breakdowns, loss ratios, and earned premium projections.

2. Dataset Overview

Key finding: The portfolio shows a total loss ratio of approximately **2.06x** — meaning claims far exceed premiums. This is a deliberate outcome of the simulation design where premiums are set at INR 100/year and claims at INR 10,000. The **1-year tenure policies** emerge as relatively the most profitable segment.

INR 24.01 Cr	INR 39.46 Cr	INR 10.00 Cr	INR 6.59 Cr
Total Premium Collected (2024)	Total Claims Cost (2025)	Total Claims Cost (2026 Jan–Feb)	Earned Premium (up to Feb 28, 2026)

2.1 Policy Sales Data

1,000,000 policies were simulated for calendar year 2024. Purchase dates were uniformly distributed across all 366 days. Each policy begins 365 days after purchase. Premiums are INR 100 per year of tenure; all vehicles are valued at INR 1,00,000.

Field	Description	Value / Logic
Customer_ID / Vehicle_ID	Unique identifier	1 to 1,000,000
Vehicle_Value	Insured vehicle value	INR 1,00,000 (fixed)
Premium	Annual premium per tenure	INR 100 x Tenure Years
Policy_Purchase_Date	Date of vehicle purchase	2024-01-01 to 2024-12-31 (uniform)
Policy_Start_Date	Coverage start date	Purchase Date + 365 days
Policy_End_Date	Coverage end date	Start Date + Tenure x 365 days
Policy_Tenure	Duration in years	1Y(20%), 2Y(30%), 3Y(40%), 4Y(10%)

2.2 Claims Data

2025 Claims: 30% of vehicles purchased on the 7th, 14th, 21st, or 28th of any month (~131,018 eligible) file a claim on their policy start date. Result: **39,457 claims** at INR 10,000 each.

2026 Claims: 10% of all 4-year tenure policies file a claim distributed evenly across Jan 1–Feb 28, 2026. Result: **9,997 claims** at INR 10,000 each. A vehicle that filed a 2025 claim could also be selected if it holds a 4-year policy.

3. Analytical Query Results

Query 1 — Total Premium Collected in 2024

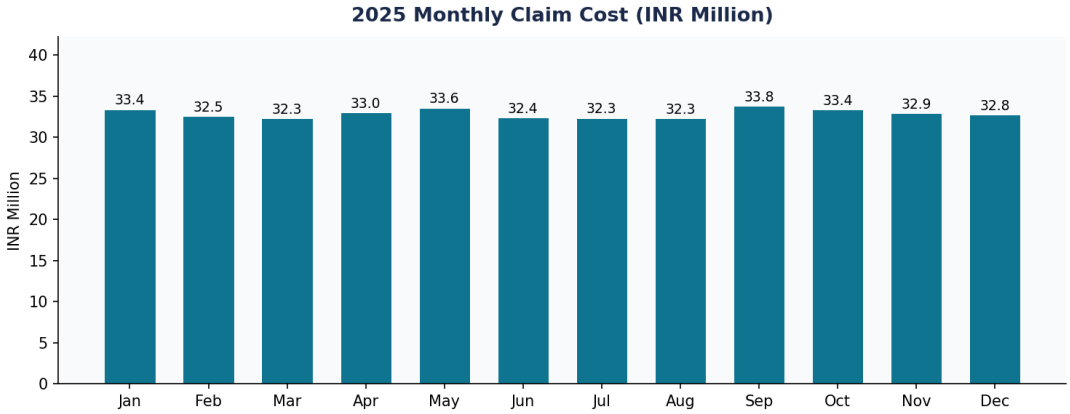
Sum of the Premium column across all 1,000,000 policies purchased in 2024.

Total Premium (2024): **INR 24,01,10,800** (≈ INR 24.01 Crore)

Query 2 — Total Claim Cost with Monthly Breakdown

2025 total: INR 39,45,70,000 (~39.46 Crore). **2026 (Jan–Feb) total:** INR 9,99,70,000 (~10.00 Crore).

Month	2025 Claim Cost (INR)	2026 Claim Cost (INR)
Jan	32,230,000	52,010,000
Feb	32,580,000	47,730,000
Mar	32,830,000	—
Apr	33,410,000	—
May	31,780,000	—
Jun	33,220,000	—
Jul	33,410,000	—
Aug	32,510,000	—
Sep	32,900,000	—
Oct	32,970,000	—
Nov	32,780,000	—
Dec	32,430,000	—
TOTAL	**39,45,70,000**	**9,99,70,000**


2025 Monthly Claim Cost (INR Million)
INR Million
0
5
10
15
20
25
30
35
40
33.4
32.5
32.3
33.0
33.6
32.4
32.3
32.3
33.8
33.4
32.9
32.8
Jan
Feb
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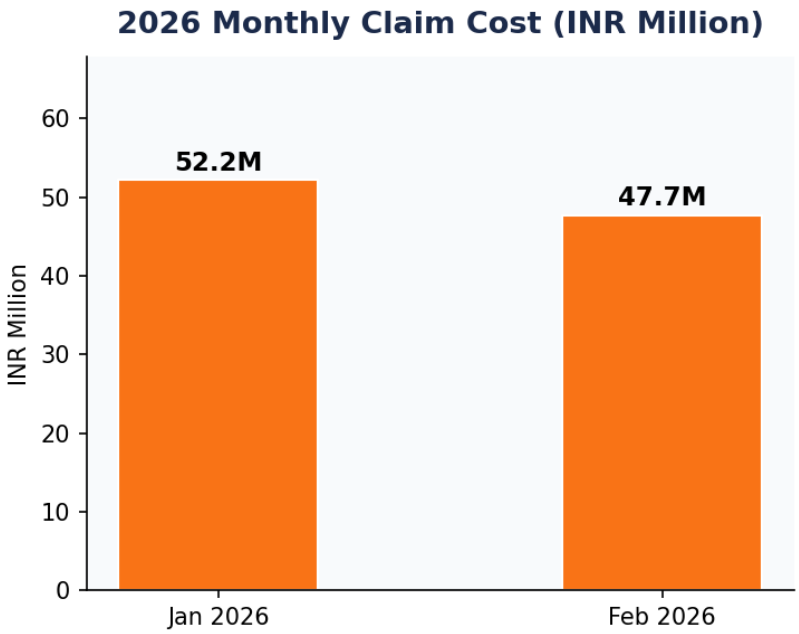
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20
30
40
50
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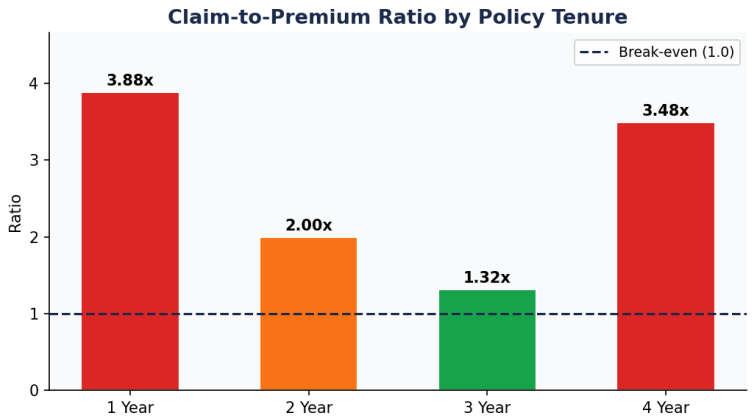
52.2M
47.7M
Jan 2026
Feb 2026

Query 3 — Claim-to-Premium Ratio by Policy Tenure

The ratio measures total claims paid per rupee of premium collected. A ratio above 1.0 means the segment is loss-making.

Tenure	Total Premium (INR)	Total Claims (INR)	Claim/Premium	Assessment
1 Year	4,80,00,000	18,61,00,000	3.88x	High Risk
2 Year	7,20,00,000	14,36,00,000	2.00x	Moderate Risk
3 Year	9,60,00,000	12,64,00,000	1.32x	Lowest Risk
4 Year	4,00,00,000	13,93,00,000	3.48x	High Risk



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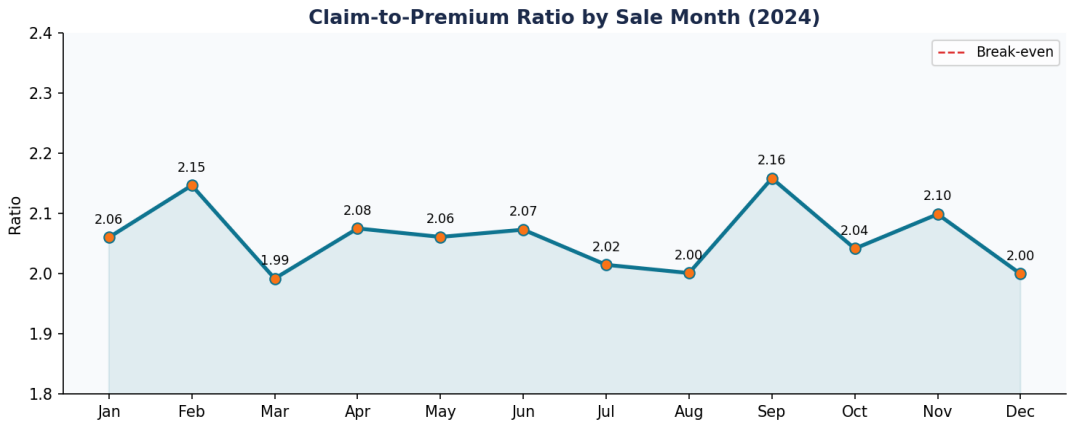

Month	Ratio	Month	Ratio	Month	Ratio
Jan	2.061x	May	2.061x	Sep	2.159x

Feb	2.147x	**Jun**	2.073x	**Oct**	2.042x
Mar	1.992x	**Jul**	2.015x	**Nov**	2.099x
Apr	2.076x	**Aug**	2.001x	**Dec**	2.000x

Query 5 — Estimated Total Potential Claim Liability

Vehicles that have not yet filed any claim are assumed to eventually file exactly one claim during remaining policy tenure.

Metric	Value
Total Policies	1,000,000
Vehicles with at least one claim	49,454
Unclaimed vehicles	950,546
Claim amount per vehicle	INR 10,000
Estimated total potential liability	**INR 9,505,460,000 (~INR 950 Crore)**

Note: This is the theoretical maximum. In practice actual liability will be lower as not all holders will claim.

Query 6 — Earned Premium & Future Premium Projection

Daily Premium = Total Premium / Total Tenure Days per policy. Earned premium is the sum of daily premiums for each active day up to Feb 28, 2026.

Metric	Value
Total Premium Collected (2024)	INR 24,01,10,800
Earned Premium up to Feb 28, 2026	INR 6,58,69,423
Unearned / Remaining Premium	INR 17,42,41,377
Remaining policy period assumed	46 months
Monthly Expected Earned Premium	**INR 37,87,856**

Note: Only ~27.4% of total premium has been earned by Feb 28, 2026, as most policies start in 2025 and extend several years forward.

4. Bonus Analysis

B1 — Most Profitable Policy Tenure

The 3-year tenure policy is the most profitable, with a claim-to-premium ratio of **1.32x** — the lowest of all segments. Key reasons:

Factor	Explanation
Largest cohort	40% of all policies; highest premium base (~INR 9.6 Cr)
Moderate claim exposure	Not targeted in 2026 claim wave; only 2025 date-based claims apply
Lower per-policy claim frequency	Only 30% of eligible 7/14/21/28 buyers in 3Y group claim
Not in 2026 wave	2026 claims exclusively target 4-year tenure policies

B3 — Overall Loss Ratio

Loss Ratio = Total Claims (2025+2026) / Total Premium = INR 49,45,40,000 / INR 24,01,10,800 = **206.0%**. A ratio above 100% means the portfolio is loss-making — expected in this simulation since claim amounts (INR 10,000) were set 100x higher than annual premium (INR 100).

B4 — Impact of 5% Annual Claim Frequency Increase

Compounding 5% annual growth on the 2025 base shows claims rising ~34% by 2030:

Year	Projected Claim Cost (INR)	Additional vs 2025	Loss Ratio
2025	39,45,70,000	—	1.64x
2026	41,43,00,000	+19,70,000	1.73x
2027	43,50,00,000	+39,60,000	1.81x
2028	45,68,00,000	+62,30,000	1.90x
2029	47,96,00,000	+82,60,000	2.00x
2030	50,36,00,000	~+10.7 Cr	2.10x

A 5% annual frequency increase means by 2030, annual claim costs are ~34% above the 2025 baseline, deepening losses further.

5. Code Review Notes

The original notebook (Assignment.ipynb) correctly implements most of the assignment. One bug was identified in Query 6:

Issue	Root Cause	Fix
KeyError: 0 when computing Earned_Days	policy_df[["Policy_End_Date"]].apply(lambda x: min(x[0], cutoff)) — when applying on a DataFrame, x[0] fails because index is the column name, not an integer.	Use policy_df.apply(lambda row: min(row["Policy_End_Date"], cutoff), axis=1) instead, operating directly on each row.

All other cells executed correctly. Dataset creation, claims simulation, and Q1–Q5 produced results consistent with the assignment specification.

6. Key Insights

INSIGHT 01 **Portfolio is Structurally Loss-Making by Design**

Total Claims / Total Premium = 206% — premiums at INR 100 vs claims at INR 10,000

The portfolio collects INR 240.1 million in premiums but pays INR 494.5 million in claims. Real-world viability requires premiums of at least INR 200–300/year per tenure year to target a 70–80% loss ratio.

Recommended Action: Immediate repricing required. Risk-adjusted premiums must reflect actual claim frequency and severity.

INSIGHT 02 **3-Year Tenure is the Most Profitable Segment**

3Y Claim/Premium Ratio = 1.32x — lowest of all tenure groups

The 3-year cohort (400,000 policies) generates the most total premium (~INR 9.6 Cr) and has the lowest claims ratio. Not targeted in the 2026 claim wave, giving it a relative structural advantage.

Recommended Action: Consider incentivising 3-year tenure uptake via discounts or bundled benefits.

INSIGHT 03 **1-Year & 4-Year Policies Are Highest Risk**

1Y ratio = 3.88x | 4Y ratio = 3.48x

1-year policies generate only INR 100 premium but face INR 10,000 exposure. 4-year policies face dual exposure — both 2025 date-based claims AND the dedicated 2026 wave exclusively targeting them.

Recommended Action: Re-underwrite short-tenure policies; add excess clauses for 4-year holders.

INSIGHT 04 **No Meaningful Seasonality in Claim Ratios**

Monthly ratios cluster between 2.00x (Dec) and 2.16x (Sep) — only 0.16x spread

Claim ratios across the 12 sale months are remarkably uniform, indicating consistent risk profiles throughout the year. No seasonal pricing adjustments are warranted based on current data.

Recommended Action: Monitor for real-world deviations; no immediate action needed.

INSIGHT 05 **Massive Latent Claim Liability Remains**

~950,546 unclaimed vehicles = potential INR 9,505 Crore future liability

Only ~4.95% of vehicles have filed claims so far. Even a 10% eventual claim rate on unclaimed vehicles would add ~INR 950 Crore in liability — far exceeding the entire premium base.

Recommended Action: Build adequate reserves. Consider reinsurance for tail risk. Use predictive models to flag high-risk unclaimed vehicles.

INSIGHT 06 **Only 27% of Premiums Earned by Feb 2026**

Earned Premium = INR 6.59 Cr of INR 24.01 Cr total (27.4%)

Since all policies start 365 days after purchase, premium earning is heavily back-loaded. The remaining INR ~17.4 Crore will be recognised over ~46 months at ~INR 37.9 Lakh per month.

Recommended Action: Maintain accurate unearned premium reserves. Cash flow planning is essential for this portfolio structure.

INSIGHT 07 **5% Annual Claim Growth Significantly Worsens Profitability**

By 2030, projected annual claims are ~34% above the 2025 baseline

Compounding 5% annual growth yields ~INR 52.9 Crore by 2030 vs INR 39.5 Crore in 2025. On an already loss-making portfolio, any frequency increase without premium increases deepens losses.

Recommended Action: Build dynamic repricing mechanisms. Model frequency assumptions into rate reviews. Consider no-claim bonuses to suppress frequency.

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Policy_Purchase_Date	Date of vehicle purchase	2024-01-01 to 2024-12-31 (uniform)
Policy_Start_Date	Coverage start date	Purchase Date + 365 days
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Policy_Tenure	Duration in years	1Y(20%), 2Y(30%), 3Y(40%), 4Y(10%)

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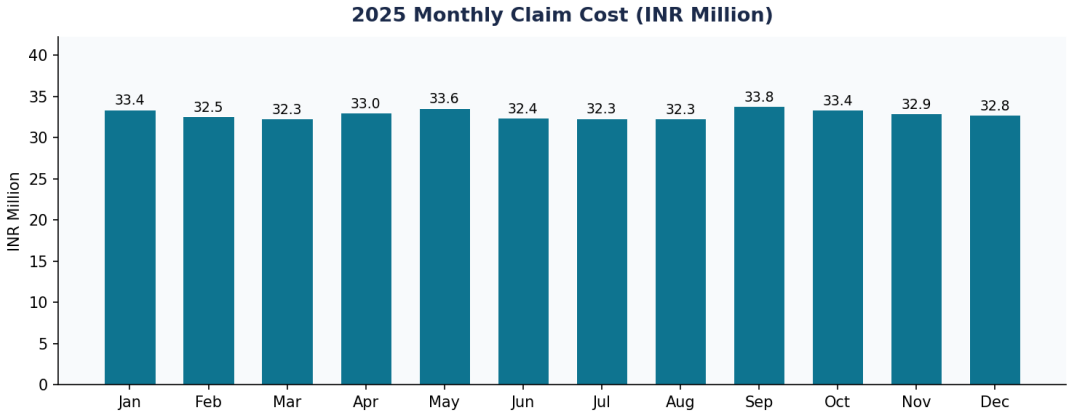

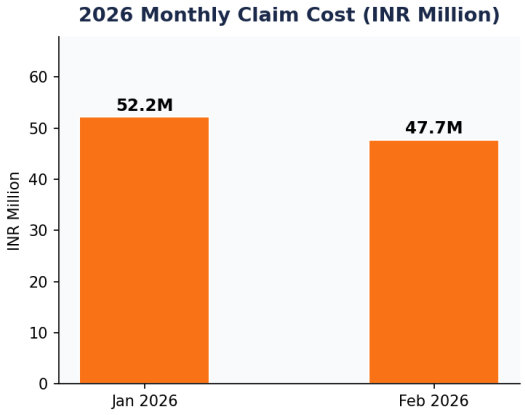
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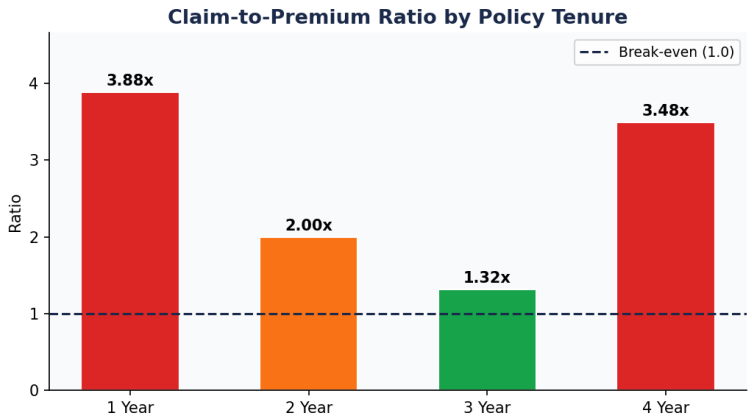

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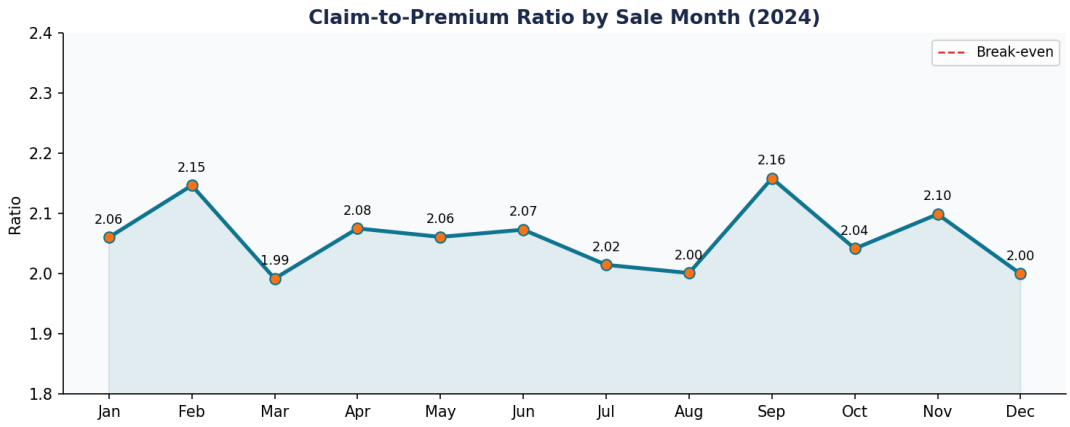

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